

Intense graFX!

Are we ready for a no-holds-barred cinema quality gaming experience? Or will the GeForce FX prove to be yet another video card with simply faster memory and processing speed?

Movies such as *Final Fantasy* and *Jurassic Park* left us gaping with their colourful tapestry of digital animation. The 'rendering farms' used to create these magic movies comprise new-age rendering supercomputers that run on advanced processing capabilities. Enchanted by such movies, the aim of every game designer and graphics hardware manager has been to bring this realism to a gaming experience. And this is exactly what the new GeForce FX aims to deliver.

While there has been no dearth of capability and programming skills to accomplish this, graphics cards were not capable of processing this vast amount of information at a pace faster than that of a slideshow. Over the years, every new card has tried to achieve realism with varying degrees of success. Hence the launch of a new graphics card usually generates more excitement in the IT community than any other hardware. Often, the revelation of the card's

specifications shows it to be merely an enhancement of the existing technology—made faster by upping clock speeds and adding more rendering pipelines, or conversely, by chopping a couple of components to make the card accessible to different market segments.

Enter the GeForce FX

The GeForce FX is built using a 0.13-micron fabrication process unlike the 0.15-micron technology used by the reigning king of graphics hill, the ATI Radeon 9700. The smaller fabrication process makes it possible for this card to be laden with 125 million transistors—compare this with the 108 million transistors used by the Xeon MP processor. While this fabrication process does offer greater density for packing in more transistors and

lower heat emission levels, it is a difficult process to implement.

This has been the primary reason for the cards delay—it was supposed to be released by December 2002, but the date has now been pushed forward to mid 2003. Also, this card complies with and even exceeds the specifications of the upcoming DirectX 9 API—while DirectX

9 will allow for a far greater effects and capabilities for tomorrow's games and applications, the GeForce FX will be able to take advantage of future graphics APIs as well. Therefore, when tomorrow's games and applications support these new capabilities, only cards such as the GeForce FX will be able to take full advantage of their visual splendour.

Let's take a look at some of the newer tech-

info

The pixel shader in the GeForce FX can process 51 billion floating point operations per second. This...

- Can render over a hundred Jurassic Park dinosaurs at 100 frames per second
- Has more floating point power than a Cray SV-1 supercomputer
- Is 120 times the distance from the Earth to the Moon, if converted to metres